

Physics 513 QFT HW 4

Pavel Okun

Department of Chemistry, University of Michigan, Ann Arbor USA

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CONTENTS

1. 1

2. 2

3. 3

1. 1

$$\phi(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}}{\sqrt{2E_p}} \quad (1)$$

$$\phi^\dagger(\vec{x}) = \int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}}}{\sqrt{2E_p}} \quad (2)$$

$$D(\vec{x} - \vec{y}) = \langle 0 | \phi(\vec{x}) \phi^\dagger(\vec{y}) | 0 \rangle \quad (3)$$

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \langle 0 | (a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}) (a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{y}} + b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{y}}) | 0 \rangle \quad (4)$$

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} (\langle 0 | a_{\mathbf{p}} a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{k}\cdot\vec{y} - \vec{p}\cdot\vec{x})} + \langle 0 | a_{\mathbf{p}} b_{\mathbf{k}} | 0 \rangle e^{i(\vec{p}\cdot\vec{x} + \vec{k}\cdot\vec{y})} + \langle 0 | b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{p}\cdot\vec{x} + \vec{k}\cdot\vec{y})} + \langle 0 | b_{\mathbf{p}}^\dagger b_{\mathbf{k}} | 0 \rangle e^{i(\vec{p}\cdot\vec{x} - \vec{k}\cdot\vec{y})}) \quad (5)$$

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \langle 0 | a_{\mathbf{p}} a_{\mathbf{k}}^\dagger | 0 \rangle e^{i(\vec{k}\cdot\vec{y} - \vec{p}\cdot\vec{x})} \quad (6)$$

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \langle \mathbf{p} | \mathbf{k} \rangle e^{i(\vec{k}\cdot\vec{y} - \vec{p}\cdot\vec{x})} \quad (7)$$

I use the normalization

$$\langle \mathbf{p} | \mathbf{k} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}) \quad (8)$$

$$D(\vec{x} - \vec{y}) = \int \frac{d^3p d^3k}{2(2\pi)^3 \sqrt{E_p E_k}} \delta^{(3)}(\mathbf{p} - \mathbf{k}) e^{i(\vec{k}\cdot\vec{y} - \vec{p}\cdot\vec{x})} \quad (9)$$

$$2(2\pi)^3 D(\vec{x} - \vec{y}) = \int d^3p \frac{e^{i\vec{p}\cdot(\vec{y} - \vec{x})}}{E_p} \quad (10)$$

$$\bar{D}(\vec{y} - \vec{x}) = \langle 0 | \phi^\dagger(\vec{y}) \phi(\vec{x}) | 0 \rangle \quad (11)$$

$$\bar{D}(\vec{y} - \vec{x}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} (\langle 0 | a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{y}} + \langle 0 | b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{y}}) (a_{\mathbf{p}} | 0 \rangle e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger | 0 \rangle e^{i\vec{p}\cdot\vec{x}}) \quad (12)$$

$$\bar{D}(\vec{y} - \vec{x}) = \int \frac{d^3p d^3k}{2(2\pi)^6 \sqrt{E_p E_k}} \langle \bar{\mathbf{k}} | \bar{\mathbf{p}} \rangle e^{i(\vec{p}\cdot\vec{x} - \vec{k}\cdot\vec{y})} \quad (13)$$

The bar denotes antiparticles.

$$\langle \bar{\mathbf{k}} | \bar{\mathbf{p}} \rangle = \langle \mathbf{p} | \mathbf{k} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}) \quad (14)$$

$$\bar{D}(\vec{y} - \vec{x}) = \int \frac{d^3p d^3k}{2(2\pi)^3 \sqrt{E_p E_k}} \delta^{(3)}(\mathbf{p} - \mathbf{k}) e^{i(\vec{p}\cdot\vec{x} - \vec{k}\cdot\vec{y})} \quad (15)$$

$$2(2\pi)^3 \bar{D}(\vec{y} - \vec{x}) = \int d^3p \frac{e^{i\vec{p}\cdot(\vec{x} - \vec{y})}}{E_p} \quad (16)$$

$$\bar{D}(\vec{x} - \vec{y}) = D(\vec{x} - \vec{y}) \quad (17)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = \left[\int \frac{d^3p}{(2\pi)^3} \frac{a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}}{\sqrt{2E_p}}, \int \frac{d^3k}{(2\pi)^3} \frac{a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{y}} + b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{y}}}{\sqrt{2E_k}} \right] \quad (18)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = \int \frac{d^3k d^3p}{2(2\pi)^6 \sqrt{E_p E_k}} [a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}, a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{y}} + b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{y}}] \quad (19)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = \int \frac{d^3k d^3p}{2(2\pi)^6 \sqrt{E_p E_k}} ([a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} + [a_{\mathbf{p}}, b_{\mathbf{k}}] e^{-i(\vec{p} \cdot \vec{x} + \vec{k} \cdot \vec{y})} + [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{i(\vec{k} \cdot \vec{y} + \vec{p} \cdot \vec{x})} + [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})}) \quad (20)$$

In HW 3 I worked out the commutators

$$[a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] = [b_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{k}). \quad (21)$$

All the other combinations are 0.

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = \int \frac{d^3k d^3p}{2(2\pi)^6 \sqrt{E_p E_k}} ([a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} + [a_{\mathbf{p}}, b_{\mathbf{k}}] e^{-i(\vec{p} \cdot \vec{x} + \vec{k} \cdot \vec{y})} + [b_{\mathbf{p}}^\dagger, a_{\mathbf{k}}^\dagger] e^{i(\vec{k} \cdot \vec{y} + \vec{p} \cdot \vec{x})} + [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})}) \quad (22)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = \int \frac{d^3k d^3p}{2(2\pi)^6 \sqrt{E_p E_k}} ([a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} + [b_{\mathbf{p}}^\dagger, b_{\mathbf{k}}] e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})}) \quad (23)$$

$$2(2\pi)^3 [\phi(\vec{x}), \phi^\dagger(\vec{y})] = \int \frac{d^3k d^3p}{\sqrt{E_p E_k}} \delta^{(3)}(\mathbf{p} - \mathbf{k}) [e^{i(\vec{k} \cdot \vec{y} - \vec{p} \cdot \vec{x})} - e^{i(\vec{p} \cdot \vec{x} - \vec{k} \cdot \vec{y})}] \quad (24)$$

$$2(2\pi)^3 [\phi(\vec{x}), \phi^\dagger(\vec{y})] = \int \frac{d^3p}{E_p} [e^{i\vec{p} \cdot (\vec{y} - \vec{x})} - e^{i\vec{p} \cdot (\vec{x} - \vec{y})}] \quad (25)$$

$$\vec{q} = \vec{y} - \vec{x} \quad (26)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] \propto \int \frac{d^3p}{E_p} (e^{i\vec{p} \cdot \vec{q}} - e^{-i\vec{p} \cdot \vec{q}}) \quad (27)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] \propto \int \frac{d^3p}{E_p} e^{i\vec{p} \cdot \vec{q}} - \int \frac{d^3p}{E_p} e^{-i\vec{p} \cdot \vec{q}} \quad (28)$$

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] \propto D(-\vec{q}) - D(\vec{q}) \quad (29)$$

For a spacelike interval between $\vec{x}$ and $\vec{y}$ I can always find a frame where the two events are simultaneous:

$$\vec{x} = (t, \mathbf{x}), \quad \vec{y} = (t, \mathbf{y}), \quad (30)$$

and

$$\vec{q} = \vec{y} - \vec{x} = (t, \mathbf{y}) - (t, \mathbf{x}) = (0, \mathbf{r}), \quad (31)$$

where $\mathbf{r} = \mathbf{y} - \mathbf{x}$.

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] \propto D[(0, -\mathbf{r})] - D[(0, \mathbf{r})] \quad (32)$$

All terms are Lorentz invariant, so I can rotate my frame: $\mathbf{r} \rightarrow -\mathbf{r}$. Thus

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] \propto D[(0, \mathbf{r})] - D[(0, \mathbf{r})] = 0. \quad (33)$$

So for spacelike separations

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = 0. \quad (34)$$

QED

2. 2

Eqs. 25 & 10 yield

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = D(\vec{x} - \vec{y}) - D(\vec{y} - \vec{x}). \quad (35)$$

Eq. 17 yields

$$[\phi(\vec{x}), \phi^\dagger(\vec{y})] = D(\vec{x} - \vec{y}) - \bar{D}(\vec{y} - \vec{x}). \quad (36)$$

QED.

I take

$$\vec{x} = (0, 0, 0, 0), \quad \vec{y} = (0, 1, 0, 0). \quad (37)$$

The interval between them is $\Delta s^2 = -1$, so they are space-like separated. Then (Eq. 10)

$$D(\vec{x} - \vec{y}) = \frac{1}{2(2\pi)^3} \int d^3p \frac{e^{i\vec{p} \cdot (0, 1, 0, 0)}}{E_p}, \quad (38)$$

$$D(\vec{x} - \vec{y}) = \frac{1}{2(2\pi)^3} \int d^3p \frac{e^{i(p^t, p^x, p^y, p^z) \cdot (0, 1, 0, 0)}}{E_p}, \quad (39)$$

$$D(\vec{x} - \vec{y}) = \frac{1}{2(2\pi)^3} \int d^3p \frac{e^{-ip^x}}{E_p}, \quad (40)$$

where

$$E_p = \sqrt{p^2 + m^2}, \quad p^2 = (p^x)^2 + (p^y)^2 + (p^z)^2. \quad (41)$$

$$2(2\pi)^3 D(\vec{x} - \vec{y}) = \int d^3p \frac{e^{-ip^x}}{\sqrt{p^2 + m^2}}. \quad (42)$$

I relabel the dummy variables as comforting spherical coordinates

$$2(2\pi)^3 D(\vec{x} - \vec{y}) = \int d^3x \frac{e^{-iz}}{\sqrt{r^2 + m^2}}. \quad (43)$$

$$2(2\pi)^3 D(\vec{x}-\vec{y}) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \int_0^\infty dr r^2 \sin\theta \frac{e^{-ir \cos\theta}}{\sqrt{r^2 + m^2}} \quad (44)$$

$$2(2\pi)^2 D(\vec{x}-\vec{y}) = \int_0^\pi d\theta \int_0^\infty dr r^2 \sin\theta \frac{\exp(-ir \cos\theta)}{\sqrt{r^2 + m^2}} \quad (45)$$

$$u = u(\theta) = \cos\theta, \quad u(0) = \cos 0 = 1, \quad u(\pi) = \cos \pi = -1, \\ -du = \sin\theta d\theta \quad (46)$$

$$2(2\pi)^2 D(\vec{x}-\vec{y}) = \int_{-1}^1 du \int_0^\infty dr r^2 \frac{\exp(-iru)}{\sqrt{r^2 + m^2}} \quad (47)$$

$$2(2\pi)^2 D(\vec{x}-\vec{y}) = \int_0^\infty dr \frac{r^2}{\sqrt{r^2 + m^2}} \int_{-1}^1 du \exp(-iru) \quad (48)$$

$$2(2\pi)^2 D(\vec{x}-\vec{y}) = \int_0^\infty dr \frac{r^2}{\sqrt{r^2 + m^2}} \\ \int_{-1}^1 du [\cos(ru) - i \sin(ru)] \quad (49)$$

$$2(2\pi)^2 D(\vec{x}-\vec{y}) = \int_0^\infty dr \frac{r^2}{\sqrt{r^2 + m^2}} \int_{-1}^1 du \cos(ru) \quad (50)$$

$$\int_{-1}^1 du \cos(ru) = \frac{2}{r} \sin r \quad (51)$$

$$(2\pi)^2 D(\vec{x}-\vec{y}) = \int_0^\infty dr \frac{r \sin r}{\sqrt{r^2 + m^2}} \quad (52)$$

This integral is not convergent, which implies that particles can't travel faster than light. But in QFT we should not expect well behaved integrals. This one can be solved with a trick

$$(2\pi)^2 D(\vec{x}-\vec{y}) = \lim_{\epsilon \rightarrow 0^+} \int_0^\infty dr \frac{r \sin r}{\sqrt{r^2 + m^2}} e^{-\epsilon r}. \quad (53)$$

For simplicity I take $m = 0$.

$$(2\pi)^2 D(\vec{x}-\vec{y}) = \lim_{\epsilon \rightarrow 0^+} \int_0^\infty dr e^{-\epsilon r} \sin r \quad (54)$$

$$(2\pi)^2 D(\vec{x}-\vec{y}) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{1 + \epsilon^2} = 1 \quad (55)$$

$$D(\vec{x}-\vec{y}) = \frac{1}{(2\pi)^2}. \quad (56)$$

Thus I found an example where a massless particle moves faster than light! This proves that particles in QFT can move faster than light, and Eqs. 34 & 36 imply

$$\bar{D}(\vec{y}-\vec{x}) = \frac{1}{(2\pi)^2}. \quad (57)$$

So antiparticles can also move faster than light. QED.

3. 3

I use my preferred notation.

$$i\mathcal{G}_F(\vec{x}-\vec{y}) = \langle 0 | T \phi(\vec{x}) \phi^\dagger(\vec{y}) | 0 \rangle \quad (58)$$

$$i\mathcal{G}_F(\vec{x}-\vec{y}) = \\ \langle 0 | \phi(\vec{x}) \phi^\dagger(\vec{y}) \Theta(x^0 - y^0) + \phi^\dagger(\vec{y}) \phi(\vec{x}) \Theta(y^0 - x^0) | 0 \rangle \quad (59)$$

$$i\mathcal{G}_F(\vec{x}-\vec{y}) = \langle 0 | \phi(\vec{x}) \phi^\dagger(\vec{y}) | 0 \rangle \Theta(x^0 - y^0) + \\ \langle 0 | \phi^\dagger(\vec{y}) \phi(\vec{x}) | 0 \rangle \Theta(y^0 - x^0) \quad (60)$$

From Sec. 1

$$D(\vec{x}-\vec{y}) = \langle 0 | \phi(\vec{x}) \phi^\dagger(\vec{y}) | 0 \rangle = \frac{1}{2(2\pi)^3} \int d^3p \frac{e^{i\vec{p} \cdot (\vec{y}-\vec{x})}}{E_p}, \\ \bar{D}(\vec{y}-\vec{x}) = \langle 0 | \phi^\dagger(\vec{y}) \phi(\vec{x}) | 0 \rangle = \frac{1}{2(2\pi)^3} \int d^3p \frac{e^{i\vec{p} \cdot (\vec{x}-\vec{y})}}{E_p}. \quad (61)$$

$$2(2\pi)^3 i\mathcal{G}_F(\vec{x}-\vec{y}) = \\ \int \frac{d^3p}{E_p} [e^{i\vec{p} \cdot (\vec{y}-\vec{x})} \Theta(x^0 - y^0) + e^{i\vec{p} \cdot (\vec{x}-\vec{y})} \Theta(y^0 - x^0)] \quad (62)$$